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Palladium-catalysed alleneamination of γ, δ-unsaturated hydrazones with propargylic acetates: access to tetrahydropyridazines bearing allenes†

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A highly efficient palladium-catalysed alleneamination of γ, δ-unsaturated hydrazones with propargylic acetates has been developed. This novel method was used to synthesize tetrahydropyridazines bearing allene derivatives, and did so in moderate to good yields. This protocol features a simple catalytic system, broad substrate scope, easy scale-up, and versatile transformations.

Tetrahydropyridazines constitute a class of six-membered N-heterocycles and represent a privileged motif in various natural products.[1] Moreover, tetrahydropyridazines and their derivatives exhibit a large range of pharmacological activities,[2,3] such as neuro-transmitter inhibitor, antimicrobial, influenza neuraminidase inhibitor, and cannabinoid CB_1 receptor activities. In addition, they play an important role in synthetic organic chemistry as versatile building blocks due to the presence of an N–N bond (Fig. 1).[4] Therefore, the development of efficient synthetic methodologies for their preparation has attracted considerable attention.

Over the past two decades, significant efforts have been devoted to developing elegant routes toward six-membered nitrogen-containing heterocycles, including hydrogenation of pyridine[5] and aza-Diels–Alder reactions (Scheme 1a).[6] Hydrazones are key structural motifs and have been widely used in drug development, materials science, and biological probes.[7] Meanwhile, the advantages of hydrazones, including their easy preparation, the presence of an N–N bond, and their role as synthetic equivalents to ketones or aldehydes enables their extensive use for synthesizing a range of N-heterocycles.[8–10] Especially, the hydrazones containing unsaturated C–C double bonds have been frequently studied for the synthesis of tetrahydropyridazines. For instance, Xiao's, Liu 's and Stuart's group independently realized the synthesis of functionalized tetrahydropyridazines through 6-*endo*-trig cyclization (Scheme 1b, left).[9] Afterwards, the Yu and Pan group developed a similar method that produced sulfonylated tetrahydropyridazines in moderate to good yields (Scheme 1b, right).[10] In addition, the Chen and Xiao group[11] in 2017 described Pd(II)-catalysed aminoarylation reactions of γ,δ-unsaturated hydrazones, providing general approaches to tetrahydropyridazines. Later, Zhang and Zhao reported an efficient Pd(0)-catalysed cyclization-coupling reaction of readily available γ,δ-alkenyl hydrazone with organohalides for the synthesis of tetrahydropyridazines (Scheme 1c).[12] Despite these achievements, additional efforts for their efficient synthesis are highly desirable.

Allenes are useful building blocks for the synthesis of pharmaceuticals, natural products, and functional materials.[13,14] A variety of methods for the efficient synthesis of different kinds of allenes have been well developed.[14] Transition-metal–catalysed reactions of propargylic alcohol derivatives with various reagents represent one of the most useful and efficient methodologies for synthesizing allenes (Scheme 1d).[15] In this context, we set out in the current work to develop an efficient palladium-catalysed alleneamination of γ,δ-unsaturated hydrazones with propargylic acetates, leading to

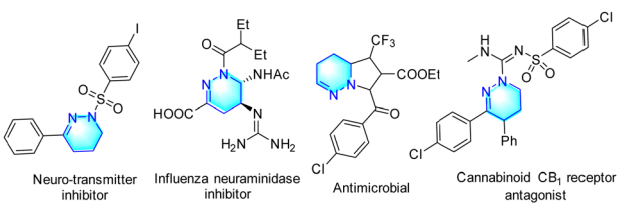
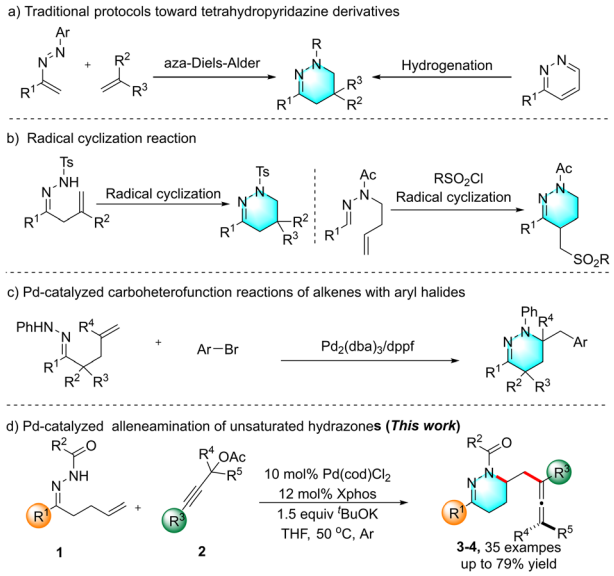


Fig. 1 Bioactive molecules bearing the tetrahydropyridazine motif.

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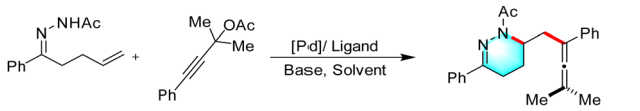
† Electronic supplementary information (ESI) available: Experimental procedures, spectroscopic data for the substrates and products. CCDC 2409294. For ESI and crystallographic data in CIF or other electronic format see DOI: https://doi.org/10.1039/d4cc06716a

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Scheme 1 Strategies for the synthesis of tetrahydropyridazines. (a) Traditional protocols toward tetrahydropyridazine derivatives. (b) Radical cyclization reaction. (c) Pd-catalyzed carboheterofunction reactions of alkenes with aryl halides. (d) Pd-catalyzed cross-coupling with nucleophilic reagent. (e) Pd-catalyzed alleneamination of unsaturated hydrazones.

Table 1 Optimization of reaction conditions[a]



Entry	Pd catalyst	Ligand	Base	Solvent	Yield[b] (%)
1	$Pd(cod)Cl_2$	PPh_3	KO^{t}Bu	THF	42
2	$Pd(cod)Cl_2$	XPhos	KO^{t}Bu	THF	73
3	$Pd(cod)Cl_2$	DpePhos	KO^{t}Bu	THF	22
4	$Pd(cod)Cl_2$	RuPhos	KO^{t}Bu	THF	72
5[c]	$Pd_2(dba)_3$	XPhos	KO^{t}Bu	THF	57
6	$Pd(OAc)_2$	XPhos	KO^{t}Bu	THF	64
7	$PdCl_2$	XPhos	KO^{t}Bu	THF	Trace
8[c]	$[Pd(allyl)Cl]_2$	XPhos	KO^{t}Bu	THF	71
9	$Pd(cod)Cl_2$	XPhos	K_2CO_3	THF	0
10	$Pd(cod)Cl_2$	XPhos	NaO^{t}Bu	THF	36
11	$Pd(cod)Cl_2$	XPhos	Cs_2CO_3	THF	3
12	$Pd(cod)Cl_2$	XPhos	KO^{t}Bu	Toluene	48
13	$Pd(cod)Cl_2$	XPhos	KO^{t}Bu	CH_3CN	Trace
14	$Pd(cod)Cl_2$	XPhos	KO^{t}Bu	Dioxane	Trace
15	$Pd(cod)Cl_2$	XPhos	KO^{t}Bu	DCE	7
16[d]	$Pd(cod)Cl_2$	XPhos	KO^{t}Bu	THF	Trace
17	$Pd(cod)Cl_2$	—	KO^{t}Bu	THF	0
18[e]	$Pd(cod)Cl_2$	XPhos	KO^{t}Bu	THF	75

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.4 mmol, 2.0 equiv.), base (0.3 mmol, 1.5 equiv.), Pd catalyst (10 mol%) and ligand (12 mol%) in THF (2.0 mL) at 50 °C under argon for 24 h. [b] Isolated yield. [c] Pd catalyst (5 mol%) was used. [d] Under air for 24 h. [e] **1a** (0.3 mmol), **2a** (0.6 mmol, 2.0 equiv.), KO^{t}Bu (0.45 mmol, 1.5 equiv.), $Pd(cod)Cl_2$ (10 mol%) and XPhos (12 mol%) in THF (2.0 mL) at 50 °C under argon for 48 h.

the formation of allene-containing tetrahydropyridazines (Scheme 1e).

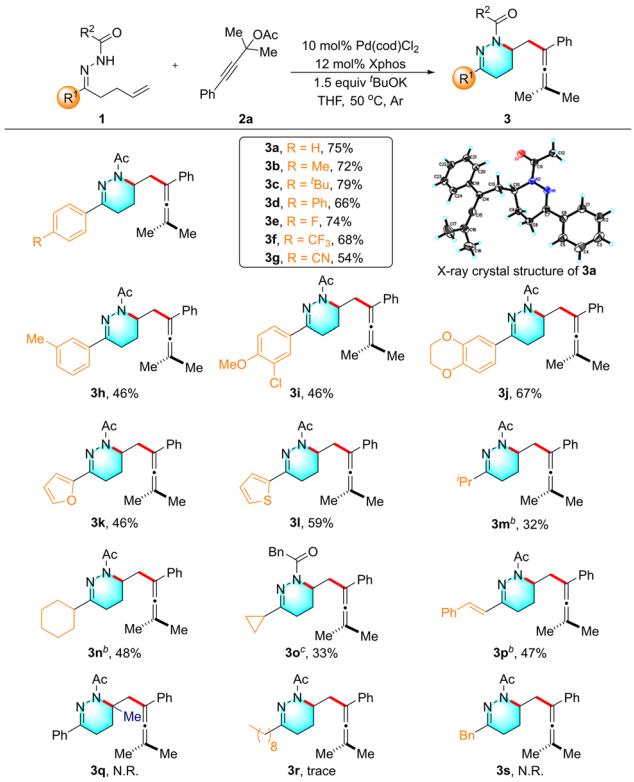
We initiated the present study with the investigation of γ, δ-unsaturated hydrazone **1a** (0.2 mmol, 1.0 equiv.) and propargylic acetate **2a** (0.4 mmol, 2.0 equiv.) as starting materials in the presence of Pd catalyst (10 mol%) and phosphine ligand (12 mol%) (Table 1). When a mixture of **1a** and **2a** in THF (2.0 mL) was treated with 10 mol% of $Pd(cod)Cl_2$ and 12 mol% of PPh_3 in the presence of 1.5 equiv. of ^{t}BuOK as a base at 50 °C under argon for 24 h, the reaction proceeded smoothly to give the desired product **3a** in 42% yield (Table 1, entry 1). Then, a series of phosphine ligands, such as XPhos, DpePhos, and RuPhos, were tested (Table 1, entries 2–4). Fortunately, the yield of **3a** improved to 73% when XPhos was used as the ligand (Table 1, entry 2). Afterwards, we tested for the effects of various palladium catalysts including $Pd_2(dba)_3$, $Pd(OAc)_2$, $PdCl_2$, and $[Pd(allyl)Cl]_2$, but no better result was observed (Table 1, entries 5–8). Replacing KO^{t}Bu with other bases such as K_2CO_3, NaO^{t}Bu, and Cs_2CO_3 led to significant decreases in the yields (Table 1, entries 9–11). Other solvents such as toluene, CH_3CN, 1, 4-dioxane, and DCE were further screened, but the product **3a** was obtained with unsatisfactory results (Table 1, entries 12–15). When the reaction was performed under an air atmosphere, only a trace amount of the product **3a** was obtained. Moreover, no product **3a** was obtained in the absence of a ligand (Table 1, entry 17). Finally, we found that the reaction efficiency did not decrease at all when the reaction was carried out on a scale of 0.3 mmol (Table 1, entry 18).

With the optimal reaction conditions in hand, we investigated the generality of γ,δ-unsaturated hydrazones **1** reacting with **2a** in the current method as shown Scheme 2. As expected, a wide range of γ,δ-unsaturated hydrazones **1** bearing different groups on the aryl ring—both electron-donating groups including methyl, phenyl, and *tert*-butyl groups and electron-withdrawing groups including fluoro, trifluoromethyl, and cyano groups—were well tolerated, yielding the corresponding allene-bearing tetrahydropyridazines **3b–3h** in moderate to good yields (46–79%). Notably, disubstituted and heterocycle-substituted substrates also reacted smoothly to deliver **3i–3l** in yields ranging from 46% to 67%. Furthermore, γ,δ-unsaturated hydrazones bearing alkyl groups, such as isopropyl, cyclohexyl, and cyclopropyl, exhibited lower reactivity, and the products **3m–3o** were produced in only 32–48% yields. In addition, styryl-substituted product **3p** was obtained, in 47% yield. Unfortunately, primaryl alkyl, benzyl, and disubstituted alkene motif substrates were not compatible, and **3q–3s** were not formed at all. Additionally, the configuration of **3a** was confirmed by the results of X-ray diffraction.[16]

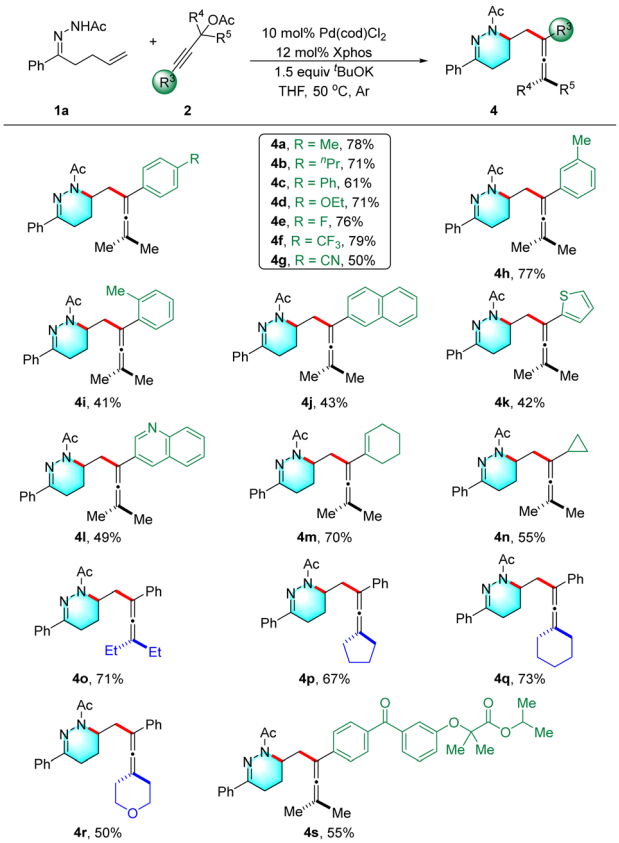
Next, the scope of propargylic acetates **2** was investigated by reacting them with **1a** (Scheme 3). Regardless of the electronic nature of the *para*-substituents on the aryl ring, functionalized propargylic acetates reacted well with hydrazone **1a** and produced a range of allene-containing tetrahydropyridazine derivatives **4a–4g** with good results (50–79% yields). When the substituents were located at a *meta*- or *ortho*-position, the desired tetrahydropyridazine products formed successfully, although steric effects could not be ignored (**4h**, 77%; **4i**, 41%). Furthermore, 2-naphthyl, cyclohexeyl, cyclopropyl, and

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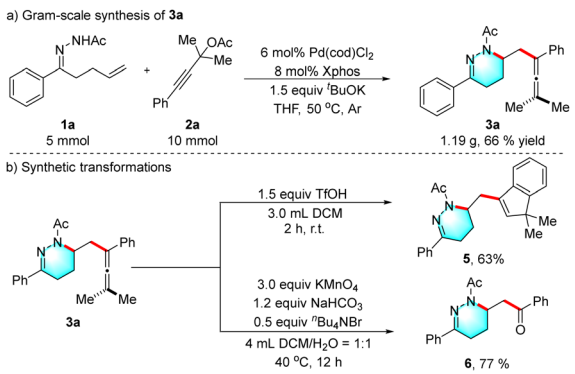
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Scheme 2 Investigation of γ,δ-unsaturated hydrazones.[a] [a] Reaction conditions: **1** (0.3 mmol), **2a** (0.6 mmol, 2 equiv.), KO^{t}Bu (0.45 mmol, 1.5 equiv.), Pd(cod)Cl$_2$ (10 mol%) and XPhos (12 mol%) in THF (2.0 mL) at 50 °C under argon for 48 h. [b] [Pd(allyl)Cl]$_2$ (5 mol%) was used, 70 °C. [c] KOH (0.45 mmol, 1.5 equiv.) and [Pd(allyl)Cl]$_2$ (5 mol%) was used, 70 °C.



Scheme 3 Investigation of propargylic acetates.[a] [a] Reaction conditions: **1a** (0.3 mmol), **2** (0.6 mmol, 2 equiv.), KO^{t}Bu (0.45 mmol, 1.5 equiv.), Pd(cod)Cl$_2$ (10 mol%) and XPhos (12 mol%) in THF (2.0 mL) at 50 °C under argon for 48 h.



Scheme 4 Gram-scale and synthetic transformations. (a) Gram-scale synthesis of **3a**. (b) Synthetic transformations.

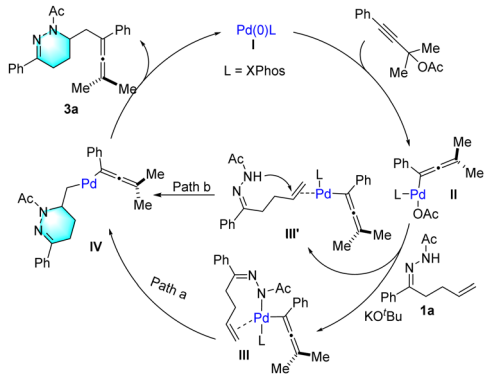
heterocycles like thiophene and quinoline were well tolerated, affording products **4j–4n** in 42–70% yields. Interestingly, the propargylic acetates bearing a cyclic system were also applicable to the current transformation, and the desired products **4p–4r** were afforded in 50–73% yields. To demonstrate the potential applications of this method, fenofibrate derivative propargylic acetate were also tested, which produced the desired coupling product **4s** in 55% yield.

To prove the practicality of this approach, a gram-scale reaction was attempted. A 5.0-mmol-scale reaction of γ, δ-unsaturated hydrazone **1a** with propargylic acetate **2a** proceeded smoothly, affording 1.19 g of **3a** with an acceptable yield of 66% (Scheme 4a). The potential applicability of the system was further illustrated by the good results of the transformations of **3a**, such as a 5-*endo* cyclization reaction with TfOH and oxidation of the allene group (Scheme 4b). In addition, the use of a KMnO$_4$ oxidation system allowed allene **3a** to be used to form ketone **6** in 77% yield.

On the basis of the previous reports and the experimental results,[17] a plausible reaction mechanism is proposed (Scheme 5). First, the oxidative addition of propargylic acetates to Pd(0) generates a Pd(II) complex **II**. Subsequently, deprotonation of **1a** and ligand exchange give intermediate **III**, which then undergoes a migratory insertion to provide complex **IV** (path a). Alternatively, coordination of alkene to Pd(II) complex **II** affords **III′**, followed by attack of alkene by the deprotonated hydrazone moiety to obtain intermediate **IV** (path b). Finally, reductive elimination of complex **IV** affords the corresponding product **3a**, and regenerates the Pd(0) catalyst for the catalytic cycle.

In conclusion, we developed an efficient palladium-catalysed alleneamination reaction, which provides an efficient and practical approach for the synthesis of allene-bearing tetrahydropyridazines in moderate to good yields. A variety of

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Scheme 5 Proposed mechanism.

readily available γ,δ-unsaturated hydrazones and functionalized propargylic acetates as starting materials demonstrated good tolerance. This protocol features a simple catalytic system, a broad substrate scope, easy scalability, and versatile transformations. Further investigation on the asymmetry of the transformation is currently underway.

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Data availability

The data supporting this article have been included in the ESI.†

Conflicts of interest

There are no conflicts to declare.

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